

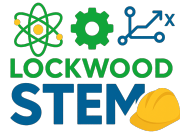
Introduction to Engineering Design

Aerospace Academy Version

Unit 1 Rubric

Introduction to Engineering Design

Course: Introduction to Engineering Design
Program: Academy of Advanced Aerospace Engineering
Brand: Lockwood STEM
Use: Student-facing and teacher grading rubrics



Unit 1

Design Process & Aerospace Missions

Mission objective, stakeholders, criteria/constraints, research, brainstorming, concept sketches, decision matrix, and design review.

Grading Rubrics and Scoring Guide

Unit 1: Design Process & Aerospace Missions

Assessment Purpose

Mission objective, stakeholders, criteria/constraints, research, brainstorming, concept sketches, decision matrix, and design review. This rubric is designed for consistent grading, feedback, and student self-checks.

Suggested 4-Point Scale

4 = Expert: exceeds expectations with clear evidence, precision, and independence. **3 = Proficient:** meets expectations with complete and accurate work. **2 = Emerging:** partially meets expectations; work is incomplete, vague, or inconsistent. **1 = Incomplete:** missing, unsafe, inaccurate, or shows limited evidence of understanding.

Unit 1 Score Summary

Category	Points Possible	Points Earned
Mission objective and problem statement	10	
Stakeholders, criteria, and constraints	10	
Research notes	10	
Brainstorming and concept development	10	
Concept sketches and annotations	15	
Trade-off analysis	10	
Decision matrix	10	
Final concept justification	10	
Design review presentation	10	
Individual reflection	5	



Lockwood STEM

Academy of Advanced Aerospace Engineering

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Unit 1 Project Rubric



Criterion	Pts	4 – Expert	3 – Proficient	2 – Emerging	1 – Incomplete	
Mission Objective & Problem Statement	10	Objective and problem statement are clear, specific, user-centered, and do not prematurely choose a solution.	Objective and problem statement are clear and mostly complete.	Objective/problem statement is vague, too broad, or solution-specific.	Missing or does not describe the engineering problem.	
Stakeholders, Criteria & Constraints	10	Identifies a complete, realistic set of stakeholders, criteria, and constraints connected to the mission.	Identifies appropriate stakeholders, criteria, and constraints with minor gaps.	Lists are incomplete, generic, or confuse criteria with constraints.	Missing or mostly inaccurate.	
Research Notes	10	Research uses multiple relevant sources and records useful features, limitations, and design implications.	Research includes relevant examples and adequate notes.	Research is limited, copied, or lacks design implications.	Missing or not connected to the mission.	
Brainstorming & Concept Development	10	Generates many varied ideas, combines concepts thoughtfully, and avoids choosing too early.	Generates several reasonable ideas and selects three for development.	Few ideas or limited variety; selection lacks reasoning.	Missing or only one idea considered.	
Concept Sketches & Annotations	15	Three clear, labeled, annotated sketches communicate function, payload protection, recovery/safety, and concerns.	Sketches are understandable with labels and some annotations.	Sketches are unclear, incomplete, or lack design intent.	Sketches are missing or not interpretable.	
Trade-Off Analysis	10	Explains realistic aerospace trade-offs such as mass, strength, cost, reliability, safety, and manufacturability.	Identifies reasonable strengths, weaknesses, and trade-offs.	Trade-offs are vague or focus only on likes/dislikes.	Missing or inaccurate.	
Decision Matrix	10	Criteria are mission-aligned; scoring is evidence-based; totals and interpretation are clear.	Matrix is complete and mostly logical.	Matrix is incomplete or scoring lacks evidence.	Missing or unusable.	
Final Concept Justification	10	Justification uses evidence from research, criteria, sketches, trade-offs, and matrix while acknowledging limitations.	Justification explains the design choice with some evidence.	Justification is mostly opinion-based or incomplete.	Missing or unsupported.	
Design Review	10	Presentation is	Presentation is clear and	Presentation is incomplete	Presentation missing or	

Design Review Presentation Rubric



Criterion	Pts	4 – Expert	3 – Proficient	2 – Emerging	1 – Incomplete
Engineering Story	20	Presentation clearly explains need, process, evidence, selected design, and next steps as one coherent engineering story.	Presentation explains main design process and final concept clearly.	Presentation is fragmented or focuses mostly on final idea.	Presentation lacks coherent explanation.
Use of Evidence	20	Uses criteria, constraints, research, sketches, matrix, data, and limitations to support claims.	Uses several pieces of evidence appropriately.	Evidence is limited, unclear, or weakly connected.	Claims are unsupported.
Visual Communication	20	Slides are clean, visual, readable, and include useful sketches/models/tables without overcrowding.	Slides are readable and include required visuals.	Slides are text-heavy, cluttered, or missing important visuals.	Slides missing or unreadable.
Team Delivery	20	All members contribute, speak professionally, stay within time, and respond to questions with engineering reasoning.	Most members contribute and delivery is clear.	Delivery is uneven, rushed, or lacks preparation.	Team is not prepared or does not present professionally.
Reflection and Next Steps	20	Clearly identifies limitations, risks, and specific next tests/redesigns.	Identifies reasonable limitations and next steps.	Next steps are vague or not connected to evidence.	No limitations or next steps.

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Rubrics

Student Self-Check Before Submission

	Requirement	Complete?
☐	Mission objective and problem statement evidence is complete and ready to be graded.	
☐	Stakeholders, criteria, and constraints evidence is complete and ready to be graded.	
☐	Research notes evidence is complete and ready to be graded.	
☐	Brainstorming and concept development evidence is complete and ready to be graded.	
☐	Concept sketches and annotations evidence is complete and ready to be graded.	
☐	Trade-off analysis evidence is complete and ready to be graded.	
☐	Decision matrix evidence is complete and ready to be graded.	
☐	Final concept justification evidence is complete and ready to be graded.	
☐	Design review presentation evidence is complete and ready to be graded.	
☐	Individual reflection evidence is complete and ready to be graded.	
☐	All pages/files are named, organized, and submitted correctly.	
☐	Reflection or revision notes are complete.	

Teacher
Use the
evidence
those b